

1. What is NumPy mainly used for?

- A) Web Development
- B) Numerical Computing
- C) Game Development
- D) Networking

2. Which statement is used to import NumPy?

- A) import numpy as np
- B) import pandas as pd
- C) import matplotlib as plt
- D) import np

3. Which function creates a NumPy array?

- A) np.array()
- B) np.make()
- C) np.create()
- D) np.list()

4. Which function creates an array of zeros?

- A) np.ones()
- B) np.zeros()
- C) np.empty()
- D) np.array()

5. Which function creates an array of ones?

- A) np.zeros()
- B) np.ones()
- C) np.arange()
- D) np.linspace()

6. Which function creates numbers from 1 to 10?

- A) np.arange()
- B) np.sum()
- C) np.mean()
- D) np.sort()

7. Which attribute shows the shape of an array?

A) `shape`

B) `size`

C) `dtype`

D) `mean`

8. Which attribute gives the total number of elements?

A) `shape`

B) `size`

C) `ndim`

D) `type`

9. Which attribute gives the number of dimensions?

A) `size`

B) `shape`

C) `ndim`

D) `length`

10. Which function changes the shape of an array?

A) `reshape()`

B) `resize()`

C) `sort()`

D) `append()`

11. Which function finds the largest value?

A) `np.max()`

B) `np.min()`

C) `np.mean()`

D) `np.sum()`

12. Which function finds the smallest value?

A) `np.max()`

B) `np.min()`

C) `np.sort()`

D) `np.sum()`

13. Which function calculates the average?

A) `np.mean()`

B) `np.sum()`

C) `np.max()`

D) `np.min()`

14. Which function adds all elements?

A) `np.sum()`

B) `np.mean()`

C) `np.max()`

D) `np.sort()`

15. Which function sorts an array?

A) `np.reshape()`

B) `np.sort()`

C) `np.append()`

D) `np.mean()`

16. Which function creates random numbers?

A) `np.random.rand()`

B) `np.zeros()`

C) `np.array()`

D) `np.mean()`

17. Which operator is used for addition?

A) `+`

B) `-`

C) `*`

D) `/`

18. Which operator is used for multiplication?

A) `+`

B) `-`

C) `*`

D) `/`

19. NumPy arrays are generally _____ than Python lists.

A) Slower

B) **Faster**

C) Equal

D) Smaller

20. Which function creates an identity matrix?

A) **np.eye()**

B) np.ones()

C) np.zeros()

D) np.arange()

21. What is Pandas mainly used for?

A) **Data Analysis**

B) Game Development

C) Networking

D) Animation

22. Which statement imports Pandas?

A) **import pandas as pd**

B) import numpy as np

C) import matplotlib.pyplot as plt

D) import pd

23. Which Pandas object is one-dimensional?

A) DataFrame

B) **Series**

C) Array

D) Matrix

24. Which Pandas object is two-dimensional?

A) List

B) Tuple

C) **DataFrame**

D) Series

25. Which function reads a CSV file?

- A) `read_csv()`
- B) `read_excel()`
- C) `open_csv()`
- D) `load_csv()`

26. Which function displays the first five rows?

- A) `head()`
- B) `tail()`
- C) `info()`
- D) `shape`

27. Which function displays the last five rows?

- A) `head()`
- B) `tail()`
- C) `info()`
- D) `describe()`

28. Which attribute gives the number of rows and columns?

- A) `shape`
- B) `size`
- C) `count`
- D) `info`

29. Which function gives information about a DataFrame?

- A) `info()`
- B) `head()`
- C) `tail()`
- D) `mean()`

30. Which function gives summary statistics?

- A) `describe()`
- B) `info()`
- C) `head()`
- D) `tail()`

31. Which function removes missing values?

A) `dropna()`

B) `fillna()`

C) `replace()`

D) `sort_values()`

32. Which function fills missing values?

A) `fillna()`

B) `dropna()`

C) `merge()`

D) `groupby()`

33. Which function sorts data?

A) `sort_values()`

B) `groupby()`

C) `merge()`

D) `head()`

34. Which function removes duplicate rows?

A) `drop_duplicates()`

B) `fillna()`

C) `merge()`

D) `head()`

35. Which function combines two DataFrames?

A) `merge()`

B) `sort()`

C) `mean()`

D) `count()`

36. Which function groups similar data?

A) `groupby()`

B) `merge()`

C) `fillna()`

D) `head()`

37. Which function counts unique values?

A) `value_counts()`

B) `shape`

C) `head()`

D) `tail()`

38. Which function saves a DataFrame as a CSV file?

A) `to_csv()`

B) `save_csv()`

C) `write_csv()`

D) `export()`

39. Which attribute returns column names?

A) `columns`

B) `shape`

C) `size`

D) `index`

40. Which function checks missing values?

A) `isnull()`

B) `merge()`

C) `sort_values()`

D) `head()`